

# PWM DC Motor Speed Controller

## Design Documentation & Technical Analysis

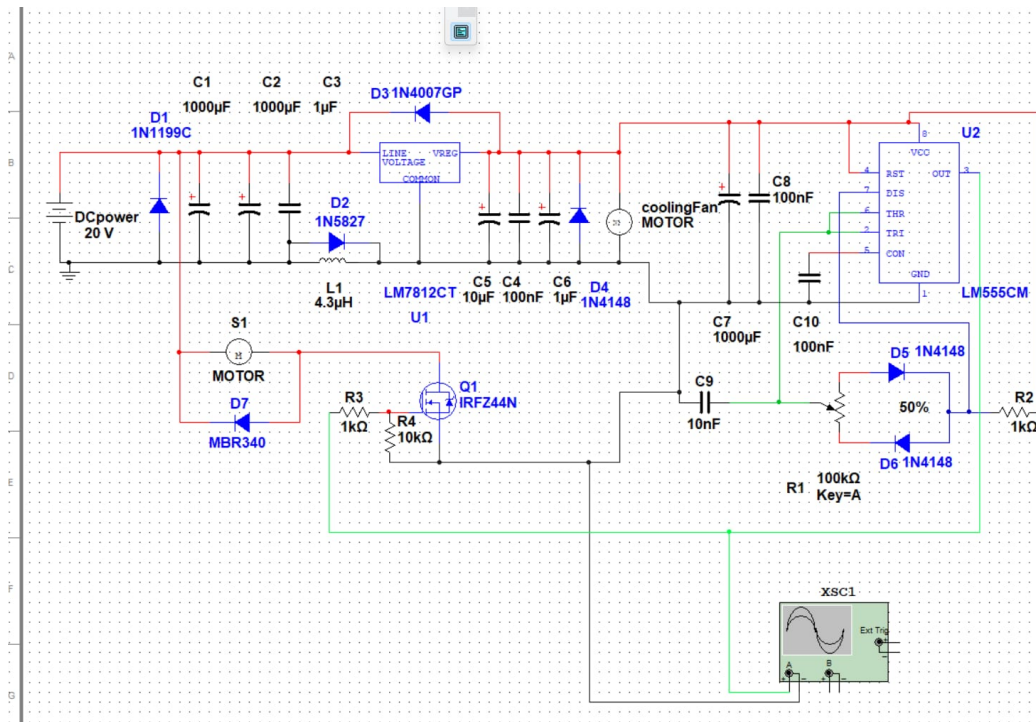


Figure 1 — Complete PWM Motor Controller Schematic

| Parameter          | Value                     |
|--------------------|---------------------------|
| Supply Voltage     | 20V DC                    |
| Logic Rail         | 12V (LM7812CT regulated)  |
| Control Method     | 555 Timer Astable PWM     |
| Power Switch       | IRFZ44N N-Channel MOSFET  |
| Flyback Protection | 40A TO-220 Schottky Diode |
| Motor Cooling      | BLDC Fan (active)         |

## 1. Project Overview

This project is a variable-speed PWM (Pulse Width Modulation) DC motor controller designed for high reliability and robust performance. The circuit uses a 555 timer IC to generate a variable duty-cycle PWM signal, which drives a power MOSFET to control the average voltage and current delivered to a DC motor. The design philosophy emphasises generous component derating, multi-stage power filtering, and complete isolation between the noisy motor power rail and the sensitive control logic rail — resulting in a circuit that outperforms many commercial off-the-shelf PWM controllers in real-world reliability.

## 2. Working Principle

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### 2.1 PWM Motor Speed Control

PWM speed control works by rapidly switching the motor supply on and off at a fixed frequency. The ratio of on-time to off-time (duty cycle) determines the average power delivered to the motor. A 50% duty cycle delivers half the supply voltage on average; 90% duty cycle delivers 90%. This switching method is far more efficient than resistive speed control because the MOSFET dissipates power only during switching transitions, not during steady-state conduction.

### 2.2 555 Timer Astable Operation

The LM555CM timer (U2) is configured in astable mode, continuously oscillating to produce the PWM waveform. The frequency is set by the RC timing network. Diodes D5 and D6 (1N4148) are placed in opposing orientations across potentiometer R1, creating separate charge and discharge paths through the timing capacitor C9. This allows the duty cycle to be varied independently of the frequency by adjusting R1 — turning the potentiometer increases duty cycle while the other direction decreases it, giving smooth and stable motor speed control.

### 2.3 MOSFET Switching Stage

The PWM output from pin 3 of the 555 timer drives the gate of Q1 (IRFZ44N), an N-channel power MOSFET configured as a low-side switch. When the gate is driven high, the MOSFET conducts and connects the motor to ground, allowing current to flow. When the gate is driven low, the MOSFET turns off and the motor current freewheels through D7. The source of Q1 is connected to ground and the motor sits between the positive 20V rail and the drain, which is the correct low-side N-channel topology.

## 3. Power Supply Architecture

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The circuit uses two completely separate power rails to isolate switching noise from the sensitive control circuitry — a key design decision that directly contributes to reliability.

### 3.1 20V Motor Rail

The 20V DC input directly powers the motor through the MOSFET switching stage. This rail carries high switching currents and generates significant electrical noise. D1 (1N1199C) provides input protection. The motor rail is filtered by C1 and C2 (1000uF bulk capacitors) for low-frequency ripple, C3 (1uF) for mid-frequency noise, and the L1/D2 LC filter network for switching transients.

### 3.2 12V Logic Rail (LM7812CT)

The LM7812CT linear voltage regulator (U1) converts the noisy 20V motor rail into a clean, stable 12V supply for the 555 timer and associated logic. Beyond voltage regulation, the LM7812CT provides approximately 60dB of Power Supply Rejection Ratio (PSRR), attenuating motor switching noise by a factor of 1000 before it can reach the control circuitry. This makes the regulator act as both a voltage converter and a noise barrier simultaneously.

## 4. Component Selection & Rationale

| Component         | Value / Part        | Reason for Selection                                                                    |
|-------------------|---------------------|-----------------------------------------------------------------------------------------|
| U2 — Timer        | LM555CM             | Push-pull output (200mA), drives MOSFET gate directly without extra buffer              |
| Q1 — MOSFET       | IRFZ44N             | Logic-level gate (fully on at 10V), 49A rated, low Rds(on) = 17.5mΩ                     |
| U1 — Regulator    | LM7812CT            | Provides stable 12V + 60dB noise rejection, isolates logic from motor rail              |
| D7 — Flyback      | 40A TO-220 Schottky | Heavily derated — handles inductive motor back-EMF spikes, TO-220 aids heat dissipation |
| D2 — Freewheeling | 1N5827 Schottky     | Fast recovery for L1 inductor freewheeling, low forward voltage drop                    |
| D3                | 1N4007GP            | Regulator output protection / reverse current blocking                                  |
| D1                | 1N1199C             | Input rail protection, ~12A rating                                                      |
| D5, D6            | 1N4148              | Separate charge/discharge paths for asymmetric duty cycle control                       |
| L1                | 4.3uH               | Filter inductor on motor rail, reduces PWM switching noise                              |
| R3                | 1kΩ                 | Series gate resistor — damps gate ringing, limits inrush current                        |
| R4                | 10kΩ                | Gate-to-source pull-down — ensures MOSFET stays off when 555 output is low              |
| R1                | 100kΩ pot           | Variable duty cycle control — adjusts motor speed                                       |
| R2                | 1kΩ                 | Timing network resistor for 555 oscillator                                              |
| Cooling Fan       | BLDC                | Active MOSFET cooling, extends Q1 current capacity to 20-30A                            |

## 5. Multi-Stage Filtering Strategy

One of the most important design decisions in this project is the extensive multi-stage filtering network. Each stage targets a different frequency range of noise, providing comprehensive suppression from low-frequency ripple to high-frequency switching transients.

### 5.1 Motor Rail Filters

| Component            | Type         | Frequency Target          | Purpose                                           |
|----------------------|--------------|---------------------------|---------------------------------------------------|
| C1, C2 (1000uF each) | Electrolytic | Low frequency (50Hz-1kHz) | Bulk energy storage, smooths input ripple         |
| C3 (1uF)             | Film/Ceramic | Mid frequency             | Bridges gap between bulk and HF caps              |
| L1 (4.3uH)           | Inductor     | High frequency            | Filters PWM switching noise on motor rail         |
| D2 (1N5827)          | Schottky     | Transient                 | Freewheeling path for L1, prevents voltage spikes |

| Component  | Type         | Frequency Target   | Purpose                        |
|------------|--------------|--------------------|--------------------------------|
| C5 (10uF)  | Electrolytic | Low-mid frequency  | Post-regulator bulk decoupling |
| C4 (100nF) | Ceramic      | High frequency     | Post-regulator HF bypass       |
| C6 (1uF)   | Film         | Mid-high frequency | Intermediate decoupling        |

### 5.2 Logic Rail Filters (555 Timer Supply)

After the LM7812CT regulator provides first-stage noise rejection, additional filtering is applied directly on the 555 timer supply pins:

- C7 (1000uF) — Large bulk capacitor providing energy reservoir and low-frequency stabilisation for the 555 VCC pin
- C8 (100nF) — High-frequency bypass capacitor directly at the 555 VCC pin, suppresses fast transients
- C9 (10nF) — Timing capacitor stability; reduces jitter on the PWM frequency
- C10 (100nF) — Pin 5 (Control Voltage) bypass; prevents noise from modulating the PWM duty cycle

This three-barrier approach (regulator + bulk cap + HF cap) means the 555 timer sees an extremely clean power supply even while the motor is drawing heavy switching currents on the same input rail.

## 6. Gate Drive Design

The gate drive circuit is carefully designed to balance switching speed, noise immunity, and component protection.

| Element                   | Value            | Role                                                                          |
|---------------------------|------------------|-------------------------------------------------------------------------------|
| 555 pin 3 output          | Push-pull, 200mA | Actively drives gate both high AND low — no floating gate                     |
| R3 (series gate resistor) | 1kΩ              | Limits peak gate current, damps LC ringing between trace inductance and gate  |
| R4 (gate-to-source)       | 10kΩ             | Pull-down ensures MOSFET is fully off during power-up and when 555 output is  |
| Logic supply              | 12V regulated    | IRFZ44N is fully enhanced at 10V — 12V gives margin, no gate driver IC needed |

With R3 = 1kΩ, the gate charges at approximately 12mA, giving a switching time of around 5 microseconds (based on IRFZ44N gate charge Qg = 63nC). This is well suited to typical motor PWM frequencies below 20kHz. The push-pull 555 output ensures the gate is actively pulled low on turn-off, preventing shoot-through and false triggering.

## 7. Protection Circuits

| Protection        | Component(s)      | Mechanism                                                                       |
|-------------------|-------------------|---------------------------------------------------------------------------------|
| Motor back-EMF    | D7 (40A Schottky) | Freewheels inductive motor current during MOSFET off-time, clamps voltage spike |
| Inductor back-EMF | D2 (1N5827)       | Freewheels L1 filter inductor current, prevents voltage spike on motor rail     |

| Protection                | Component(s)     | Mechanism                                                                       |
|---------------------------|------------------|---------------------------------------------------------------------------------|
| Regulator reverse current | D3 (1N4007)      | Blocks reverse current into LM7812CT output                                     |
| Input protection          | D1 (1N1199C)     | Protects against reverse polarity or transients on 20V input                    |
| Gate protection           | R3 + R4          | Limits inrush, ensures defined gate state at all times                          |
| MOSFET thermal            | BLDC cooling fan | Active airflow prevents MOSFET junction temperature from reaching thermal limit |

## 8. Current Handling Capability

The circuit is designed with significant current derating on all power components, which is the primary reason it outperforms commercial modules in reliability.

| Component          | Rated Current | Typical Operating  | Derating Factor       |
|--------------------|---------------|--------------------|-----------------------|
| IRFZ44N (with fan) | 49A (25C)     | 10-25A             | ~50%                  |
| D7 (40A Schottky)  | 40A           | 10-25A             | ~60%                  |
| D1 (1N1199C)       | ~12A          | 10A max            | ~80% — weakest link   |
| D2 (1N5827)        | 3A            | <1A (filter)       | >90%                  |
| Wiring             | Heavy gauge   | Full motor current | Low resistance losses |

Note: D1 (1N1199C) at approximately 12A is the most current-limited component in the signal path. For applications exceeding 10A continuous, upgrading D1 to a higher-rated device would remove this constraint. The MOSFET and flyback diode are well derated and not limiting factors.

## 9. Why This Circuit Outperforms Commercial Modules

Commercial PWM motor controllers are optimised for minimum cost. This circuit is optimised for reliability. The key differences:

| Aspect               | This Circuit                 | Typical Commercial Module    |
|----------------------|------------------------------|------------------------------|
| Flyback diode        | 40A TO-220 Schottky          | 3-5A SMD diode               |
| Power rail filtering | Multi-stage, 3 barriers      | Single bulk cap              |
| Logic isolation      | Dedicated regulated 12V rail | Direct from motor rail       |
| MOSFET cooling       | Active BLDC fan              | No cooling or small heatsink |
| Gate drive           | Proper R series + pull-down  | Direct connection or large R |
| Component derating   | 50-90% derating on all parts | 80-95% of rated limits       |
| Noise immunity       | Triple-barrier filtering     | Minimal decoupling           |

## 10. Measurement & Testing

An oscilloscope (XSC1) is included in the simulation schematic to monitor the PWM waveform at the output of the 555 timer. This allows direct measurement of:

- PWM frequency — set by R1, R2, and C9
- Duty cycle — variable via potentiometer R1 using the D5/D6 asymmetric charge/discharge path
- Waveform stability — confirms filtering is effective and 555 supply is clean
- Rise and fall times — verifies gate drive performance through R3

## 11. Summary

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This PWM DC motor speed controller represents a well-engineered, robust design that applies professional-grade techniques to a discrete component circuit. The key strengths of the design are:

- Complete isolation of noisy motor rail from sensitive 555 timer logic via LM7812CT + multi-stage filtering
- Generous component derating on all power-handling devices — particularly the 40A flyback diode
- Active MOSFET cooling extending the practical current capacity significantly
- Proper gate drive with series resistance damping and pull-down for defined switching states
- Variable duty cycle PWM using the classic D5/D6 diode trick on the 555 astable timing network
- Multiple layers of inductive flyback protection preventing voltage spikes from damaging components

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